

BookKeep Buddy

AI Bookkeeping & Operations Agent for Independent Restaurants

AI Vibe Coding Competition

June 13, 2026

book-keeping-buddy.vercel.app

github.com/mackczeller/Book-Keeping-Buddy

THE PROBLEM

The number one problem with running a restaurant is thin profit margins due to high varying food and labor costs making it almost impossible to track from data from multiple sources. Which can lead a visually flourishing restaurant to go under. Book-keep-buddy connects data from P.O.S system, quickbooks, bank statements, email, and any other data source used to run a restaurant, to help a restaurant know exactly what is going on behind the scenes.

THE SOLUTION

- **Chat tab** — The face of the book-keep-buddy designed to provide a sophisticated answer on any aspect of your restaurant. EX:(Why did my profit margin drop this week even though revenue was up?)
- **Daily Report tab** — Your daily rundown of everything that happened in a days of work with real time suggestions and alerts that help the restaurant catch a problem before it turn into a habit.
- **Dashboard tab** — Shows your most crucial statistics, their change week over week, and can even set a threshold of what ideal labor margins is.
- **Approvals tab** — With access to all this data, BookKeep Buddy prepares your Quickbooks, manages payroll, and W-9s all at the approval and discretion of the owner.

HOW AI IS USED

- **Claude API (claude-sonnet-4-5)** — Claude is the brain reading all of your data and giving you the answers to your problems
- **System prompt vs. user prompt** — BookKeep Buddy is structured to a well designed system prompt, giving Claude its own personal identity as your BookKeep Buddy. The prompt allows it to handle new and different data everyday so it reports on what actually happened instead of generalizations.
- **"Suggest and report, never act"** — One of these prompts is to never act to ensure the owner of BookKeep Buddy always has the final say in suggestions and Quickbooks updates. The AI advises, the owner decides.

TECH STACK

Frontend	React + Tailwind CSS → deployed on Vercel
Backend	Python + FastAPI → deployed on Render
AI	Claude API — claude-sonnet-4-5
Database	Supabase (5 tables)
Data	6 mock JSON files mirroring Toast POS / QuickBooks shape

KEY TECHNICAL DECISIONS

- *I built one tab for each function of the agent. Making it simpler to design, improve, and debug; leading to no one point of failure.*
- *By using mock JSON data instead of real time Toast and Quickbooks api data I was able to design and make the ai perfect before tackling real time data.*

WHAT I LEARNED

- *I learned how to properly prompt and code AI to integrate it into everyday life like BookKeep Buddy*
- *I picked up that if you are going to design a product, talk to your target audience. Tailor it to specifically what they say and what they want.*

"BookKeep Buddy watches your Toast, your invoices, and your fridge — it texts you when to send staff home, tells you when a dish stops making money, and catches theft before it becomes a habit. At the end of the day, approve your books in one click."

